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MILITARY STANDARD

INTEROPERABILITY STANDARD FOR CONNECTIONLESS DATA TRANSFER APPLICATION LAYER STANDARD



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Foreword

This military standard is approved for use by all Departments and Agencies of the Department of Defense (DOD).

Beneficial comments (recommendations, additions, deletions) and any pertinent data that may be of use in improving this MIL-STD should be addressed to the:

Joint Interoperability and Engineering Organization (JIEO)
ATTN: JEBBD
Fort Monmouth, New Jersey 07703-5613

by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this MIL-STD or by memorandum.

This MIL-STD is produced by the DCPS Technical Management Panel (DTMP). The MIL-STD-2045 document series was established within the DCPS Standardization Area to allow for the enhancement of commercial standards or the development of standards that are unique to DOD.

The MIL-STD-2045-10000 series, MIL-STD-2045-10000 to MIL-STD-2045-19999 inclusive, is used to describe how DOD implements commercial, international, national, federal, or military standards within the functional profile concept to provide required network services. The Profiles for Open Systems Internetworking Technologies (POSIT) with DOD enhancements will serve as the base for developing the 10000 series. POSIT has superseded the Government Open Systems Interconnection Profile (GOSIP). Unique military standards will be used only when necessary.

The MIL-STD-2045-20000 series, MIL-STD-2045-20000 to MIL-STD-2045-29999 inclusive, is used to describe DOD enhancements and extensions to existing commercial, international, national, or federal standards.

The MIL-STD-2045-30000 series, MIL-STD-2045-30000 to MIL-STD-2045-39999 inclusive, is used to describe protocols and services unique to DOD that will not be supported by commercial, international, national, or federal standards.

The MIL-STD-2045-40000 series, MIL-STD-2045-40000 to MIL-STD-2045-49999 inclusive, is used to document interim standards. Interim standards document protocols and services needed by DOD until tnese protocols and services are described in either GOSIP (or POSIT) or in a MIL-STD-2045-20000 or -30000 series standard.

The MIL-STD-2045-50000 series, MIL-STD-2045-50000 to MIL-STD-2045-59999 inclusive, is used to describe how DOD will implement commercial, international, national, federal, or military standards within the functional profile concept to provide necessary interim network services. The POSIT will serve as the base for developing the 50000 series, with DOD enhancements. Unique military standards will be used only when necessary.

Specific details and instructions for establishing a MIL-STD-2045 document, as well as profile development guidelines, are documented in MIL-HDBK-829. DTMP Working Groups shall be responsible for DSP development and informal service and agency coordination; the DTMP Plenary shall be responsible for final review and approval.

This MIL-STD does not supersede the scope of Allied Communication Publication (ACP) 123 with US SUPP-1. ACP 123 with US SUPP-1 addresses message handling communications protocol and procedures for the exchange of military messages.

The Preparing Activity for this standard is the Data Communication Protocol standards Technical

Management Panel (DTMP). The custodians for the document are identified in the Defense Standardization Program, "Standardization Directory (SD-1)" and are classified in the Federal Supply Classification (FSC) system under Data Communication Protocol Standards (DCPS). Additional information can be obtained from:

Joint Interoperability and Engineering Organization
ATTN: DTMP Chairman, JEBBD
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CONTENTS

1.	**SCOPE**	1
1.1	Purpose.	1
1.2	Scope	1
1.3	Application guidance	1
1.4	System standards and design	1
2.	**APPLICABLE DOCUMENTS**	1
2.1	Government documents	1
2.1.1	Specifications, standards, and handbooks	1
2.1.2	Other Government documents, drawings, and publications	2
2.2	Non-Government documents	2
2.3	Order of precedence.	2
3.	**DEFINITIONS**	2
3.1	Definitions of terms	2
3.2	Abbreviations and acronyms	3
4.	**GENERAL REQUIREMENTS**	3
4.1	Application Layer Users	3
4.2	Interoperability.	4
4.3	Framework	4
4.4	Application-layer services provided	4
5.	**SPECIFIC REQUIREMENTS**	4
5.1	Application layer	4
5.2	Application protocol data unit	5
5.3	Application Heade	5
5.4	Application header formatting.	
5.5	Syntax.	5
5.5.1	Field presence indicator	5
5.5.2	Field recurrence indicator	5
5.5.3	Group presence indicator	5
5.5.4	Group recurrence indicator.	5
5.5.5	End-of-literal field marker	6
5.5.6	Data-field construction procedures	6
5.5.6.1	ASCII data element.	6
5.5.6.2	Binary data element	6
5.5.6.3	Header format notations.	7
5.6	Application header fields	10
5.6.1	Originator, Recipient, and Information Addressee fields.	10
5.6.2	User Message Format field.	10
5.6.3	Functional Area Designator field	11
5.6.4	Message Number field	11
5.6.5	Message Subtype Number field	11
5.6.6	Message Size field.	11
5.6.7	Operation Indicator field.	11
5.6.8	Retransmit Indicator field.	12
5.6.9	Message Precedence field.	12
5.6.10	Security Classification field.	12
5.6.11	Control and Release Marking field.	13
5.6.12	Originator Date and Time Group (DTG).	13
5.6.13	Time Perishability DTG.	13
5.6.14	Machine Acknowledge Request Indicator field.	13

5.6.15	Operator Acknowledge Request Indicator field.	13
5.6.16	Operator Reply Request Indicator field.	13
5.6.17	Message Acknowledgment DTG.	13
5.6.18	Receipt/Compliance field.	13
5.6.19	Cannot Comply (CANTCO) Reason field	14
5.6.20	Cannot Process (CANTPRO) Reason field.	14
5.6.21	Reply Amplification field.	15
5.7	Application header formatting rules and construction procedures.	15
5.7.1	Case and conditionality statement syntax.	15
5.7.1.1	Logical operators.	15
5.7.1.2	Application.	15
5.7.1.3	Reserved words	15
5.7.1.4	Cases	16
5.7.1.5	Conditions	16
5.7.1.6	Defaults	16
5.7.1.7	Expected response	16
5.7.1.8	Special considerations	17
5.7.1.9	Application header receipt	17
5.7.2	Cases and conditions for the application header.	17
5.7.2.1	Cases.	17
5.7.2.2	Conditions.	17
5.7.2.3	Defaults.	18
5.7.3	Expected Response.	18
5.7.3.1	Machine Acknowledge Requested	18
5.7.3.2	Operator Acknowledge Requested	19
5.7.3.3	Operator Reply Requested	19
5.7.4	Special Considerations.	19
5.7.4.1	Perishable Data Check	19
5.7.5	User data.	19
5.7.5.1	Message acknowledgments	19
5.7.5.2	Acknowledgment header format	19
5.7.5.3	Message accountability.	20
5.7.5.4	Message retransmission.	20
5.8	Processing Factors.	20
5.8.1	Application process.	20
5.8.2	Message formats	20
6.	**NOTES**	21
6.1	Subject term (key word) listing.	21
6.2	Issue of the DOD Index of Specifications and Standards	21
Appendix A	Application Header Fields and Codes for Variable Message Formats	23
A.1	General	23
A.1.1	Scope	23
A.1.2	Application	23
A.2	Applicable Documents	23
A.3	Codeword Tables	23
A.3.1	Unit Reference Number codewords	23
A.3.2	Functional Area Designator codewords	23
A.3.3	Message Number codewords	24
A.3.4	Message Subtype Number codewords	24
A.3.5	CANTCO Reason codewords	24
A.3.6	CANTPRO Reason codewords	24
Concluding Material		27

TABLES

TABLE I. Application Header 8
TABLE II. User Message Format codes 11
TABLE III. Operation indicator codewords 12
TABLE IV. Message Precedence codewords 12
TABLE V. Security Classification codewords 13
TABLE VI. Date and Time Group codewords 13
TABLE VII. Receipt/Compliance(R/C) codes 14
TABLE VIII. Logical Operator Definitions 15
TABLE A-1. Functional Area Designator codewords 24
TABLE A-2. CANTCO Reason codewords 24
TABLE A-3. CANTPRO Reason codewords 25

FIGURES

FIGURE 1. Combat net radio functional reference model 4
FIGURE 2. Application protocol data unit structure 5
FIGURE 3. Application protocol data field bit order (example) 6

1 SCOPE

1.1 Purpose. This MIL-STD presents the minimum essential technical parameters in the form of a mandatory system standard and optional design objectives for interoperability and compatibility among digital message transfer devices (DMTDs), between DMTDs and applicable command, control, communications, computers, and intelligence (C4I) systems and among C4I systems using digital data for information transfer over limited bandwidth communication channels.

1.2 Scope. This MIL-STD addresses part of the communications protocol and procedures for the exchange of digital data among DMTDs, between DMTDs and C4I systems, and among C4I systems participating in inter- and intra-Service tactical networks. The material is presented in the context of the Open Systems Interconnection (OSI), as documented in national and international standards.

1.3 Application guidance. This MIL-STD applies to the design, construction, and development of new equipment and systems, and to the retrofit of existing equipment and systems.

1.4 System standards and design. The parameters and other requirements specified in this MIL-STD are mandatory if the word *shall* is used in connection with the parameter value or requirement under consideration. Non mandatory objectives are indicated in parentheses after a standardized parameter value or by the word *should* in connection with the parameter value or requirement under consideration.

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this MIL-STD to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the current issue of the DOD Index of Specifications and Standards (DODISS) and supplements thereto, cited in the solicitation (see 6.2).

STANDARDS:

FEDERAL:

FED-STD-1037	*Glossary of Telecommunication Terms*

MILITARY:

MIL-STD-188-220A	*Interoperability Standard for Digital Message Transfer Device Subsystems*
MIL-STD-2045-14502-1A	*Information Technology - DOD Standardized Profile - Internet Transport Profile for DOD Communications - Part 1: Transport and Internet Services*
MIL-STD-2045-14502-6A	*Information Technology - DOD Standardized Profile - Internet Transport Profile for DOD Communications - Part 6: Combat Net Radio*

[Unless otherwise indicated, copies of federal and military standards are available from the Naval Publications and Forms Center, ATTN: NPODS, 5801 Tabor Avenue, Philadelphia, PA, 19120-5099.]

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this MIL-STD to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DIA Manual 65-19 *Standard Security Markings*

Joint Interoperability of Tactical Command and Control Systems, Variable Message Format Technical Interface Design Plan (Test Edition) (VMF TIDP TE)

VOL I *VMF Overview*

VOL II *Data Element Dictionary*

VOL III *K-Series Message Formats*

Variable Message Format Interface Operating Procedures (VMF IOP)

[These documents are available from the Joint Interoperability and Engineering Organization (JIEO), Center for Standards, Information Standards Directorate (TBCA), 10701 Parkridge Blvd. Reston, VA 22091-4398.]

2.2 Non-Government documents. The following documents form a part of this MIL-STD to the extent specified herein. Unless otherwise specified, the issues of the documents that are DOD-adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

INTERNATIONAL ORGANIZATION for STANDARDIZATION (ISO):

ISO 7498	*Information Processing Systems -- Open Systems Interconnection -- Basic Reference Model, 1984.*
ISO 7498 AD 1	*Information Processing Systems -- Open Systems Interconnection -- Basic Reference Model -- Addendum 1: Connectionless-mode Transmission, 1987.*

[ISO standards are available from the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.]

2.3 Order of precedence. In the event of a conflict between the text of this MIL-STD and the references cited herein, the text of this MIL-STD takes precedence. Nothing in this MIL-STD, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. DEFINITIONS

3.1 Definitions of terms. Definitions of terms used in this MIL-STD are specified in FED-STD-1037.

3.2 Abbreviations and acronyms. Abbreviations and acronyms used in this MIL-STD are defined below. In addition, those listed in the current edition of FED-STD-1037 that are pertinent to standards referenced by this document have been included for the convenience of the reader.

ASCII	American Standard Code for Information Interchange
C	consecutive repeatable indicator
C4I	command, control, communications, computers, and intelligence
CANTCO	cannot comply
CANTPRO	cannot process
CNR	combat net radio
DFI	data field identifier
DISA	Defense Information Systems Agency
DL	data-link layer
DMTD	digital message transfer device
DOD	Department of Defense
DODISS	Department of Defense Index of Specifications and Standards
DUI	data use identifier
FED-STD	federal standard
FIPS	federal information processing standard
FPI	field presence indicator
FRI	field recurrence indicator
GPI	group presence indicator
GRI	group recurrence indicator
HAVCO	have complied
I	information; iterative repeatable indicator
IP	Internet Protocol
ISO	International Organization for Standardization
JIEO	Joint Interoperability and Engineering Organization
LSB	least significant bit
M	mandatory
MIL-STD	military standard
MR	machine receipt
MSB	most significant bit
(*n*)	repeatability factor
O	optional
OPRACK	operator acknowledge
ORR	operator response required
OSI	Open Systems Interconnection
R	repeatable field indicator
R/C	receipt/compliance
RI	routing indicator
Rn	repeatable field within group n
TCP	Transmission Control Protocol
UDP	user datagram protocol
VMF	variable message format
WILCO	will comply

4. GENERAL REQUIREMENTS

4.1 Application Layer Users. In the context of this MIL-STD, the user of the application layer is the application process that requires the communications services to effect information exchange (the transfer of digital data) between end systems.

4.2 Interoperability. Interoperability of the application entity between end systems shall be achieved by implementing the application layer protocol specified in this MIL-STD. This standard defines the minimum essential data communications parameters and protocol conventions that are necessary to support the handling and exchange of single messages or concatenated messages [a series of messages that are combined together in a single user data block for delivery to the same destination(s)] over subnetworks and point-to-point links.

4.3 Framework. The application layer protocol specified in this MIL-STD shall support the layers of the functional reference model depicted in Figure 1. This functional reference model in Figure 1 is based on the ISO 7498 OSI seven-layer model and is for reference only. The shaded portion of Figure 1 contains the connectionless application layer format that is the subject of this MIL-STD.

OSI	CNR ENVIRONMENT	SUPPORTING MIL-STD PROFILE
Application Layer	Simplified Data Transfer Protocol	None
Presentation Layer*		
Session Layer*		
Transport Layer	Segmentation/Reassembly UDP/TCP	MIL-STD 2045-14502-1A
Network Layer	IP	MIL-STD 2045-14502-1A
Link Layer	Intranet/Link Layer	MIL-STD 2045-14502-6A
Physical Layer	Physical Layer	MIL-STD 2045-14502-6A

* NOTE: These OSI-equivalent layer functions are accomplished by other means in CNR.

Figure 1. Combat net radio functional reference model

4.4 Application-layer services provided. The application layer protocol shall provide the following services to the application process in order to facilitate the reliable exchange and distribution of messages of data between end user systems:

a. identification of intended communications partners by name;
b. identification of privacy/security mechanisms required;
c. passing of quality-of-service parameters (performance and non-performance parameters);
d. synchronization of cooperating application processes;
e. message handling (distribution, receipting, and monitoring)
f. identification of constraints on data syntax (character sets, data structure); and
g. message or data transfer via connectionless operation.

5. SPECIFIC REQUIREMENTS

5.1 Application layer. The application layer shall provide the simplified message-handling protocol.

5.2 Application protocol data unit. The application protocol data unit shall be composed of an application header and user data, as shown in Figure 2.

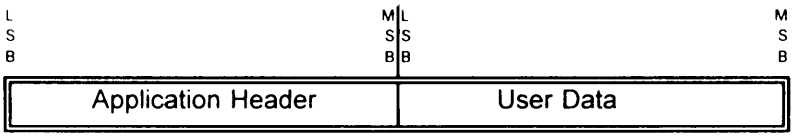


Figure 2. Application protocol data unit structure

5.3 Application Header. The application header shall consist of the fields shown in Table I. The application header may contain two categories of fields, mandatory (M) and conditional (C). A conditional field is dependent upon the presence or absence of other fields. The order of fields shall follow that shown in Table I.

5.4 Application header formatting. The application header shall use a variable format syntax and format structure . The syntax and formatting procedures are defined below.

5.5 Syntax. The application header consists of an ordered collection of bits (ones and zeros). A group is a combination of two or more related fields designated as a group. There are two types of groups, "G" groups and "R" groups. A "G" group is a combination of related fields. An "R" group is a repeatable combination of related fields. Presence and recurrence indicators as defined below shall be allowed in groups. The following syntax fields shall be used in the selection of fields to be transmitted:

a. Field Presence Indicators (FPIs). An FPI is a one bit field used to indicate the presence or absence of the following field.

b. Field Recurrence Indicators (FRIs). An FRI is a one bit field used to indicate the repeatability of a field.

c. Group Presence Indicators (GPIs). A GPI is a one bit field used to indicate the presence or absence of the following group.

d. Group Recurrence Indicators (GRIs). A GRI is a one bit field used to indicate the repeatability of a group.

5.5.1 Field presence indicator. The FPIs are used to indicate the presence (FPI = 1) or absence (FPI = 0) of the following field and are not used for mandatory fields or single bit fields. These indicators are transparent to the user, allowing the user to send only those fields containing information when use of those fields is not mandatory.

5.5.2 Field recurrence indicator. Fields may be designated as repeatable through a 1-bit FRI. If a field is preceded by an FPI, FPI = 1 shall precede the first occurrence of the FRI and is not present for following repetitions. If the FPI = 0, neither the FRI not the field is present in the application header. An FRI = 1 indicates the recurrence of the field after this iteration. An FRI = 0 indicates the field will not occur after this iteration.

5.5.3 Group presence indicator. A group is a combination of related fields. FPIs, FRIs, GPIs, and GRIs shall be allowed in groups. If a group is preceded by a GPI, then the GPI indicates the presence (GPI = 1) or absence (GPI = 0) of the group.

5.5.4 Group recurrence indicator. An "R" group is repeatable and shall be preceded by a GRI. A "G" group is not repeatable and shall not be preceded by a GRI. If an "R" group is preceded by a

GPI, GPI = 1 shall precede the first occurrence of the GRI and is not present for following repetitions. If the GPI = 0, neither the GRI nor the group is present in the application header. A GRI = 1 indicates the recurrence of the group after this iteration. A GRI = 0 indicates the group will not occur after this iteration.

5.5.5 End-of-literal field marker. The end-of-literal field marker, a 7-bit ASCII DELETE character (1111111), is used to indicate the end of free-text, character-oriented, literal fields only. The maximum literal field size is specified for each such field in Table I. Either the end-of-the literal field processing software shall be capable of recognizing both conditions.

5.5.6 Data-field construction procedures. The following construction procedures prescribe the sequence in which the application header fields are linearly joined before passing data to the next lower protocol layer. The header is constructed with elemental data fields ordered as specified in this standard. The data elements for the application header are as specified in this standard. There are two representations for data elements: 7-bit ASCII characters and binary numbers. All fields shall be joined LSB first. The LSB of the first data field or field/group indicator shall be LSB-justified within the first byte of the message buffer. The LSB of each successive data field shall be concatenated to the MSB of the preceding data field. Figure 3 uses the first few fields of the application header (from Table I) as an example of the data field bit order. The figure is interpreted as follows:

	BIT NO.	FIELD NAME	VALUE/CODE	MEANING
a.	1	GPI for Originator Address Group	0	NOT PRESENT
b.	2	GPI for Recipient Address Group	0	NOT PRESENT
c.	3	GPI for Information Address Group	0	NOT PRESENT
d.	4	GRI for Message Handling Group	0	NOT REPEATABLE
e.	5 - 8	User Message Format	2	VMF K-Series
f.	9 - 12	Functional Area Designator	7	COMBAT SERVICE SUPPORT
g.	13 - 19	Message Number	1	MEDICAL EVACUATION REQUEST

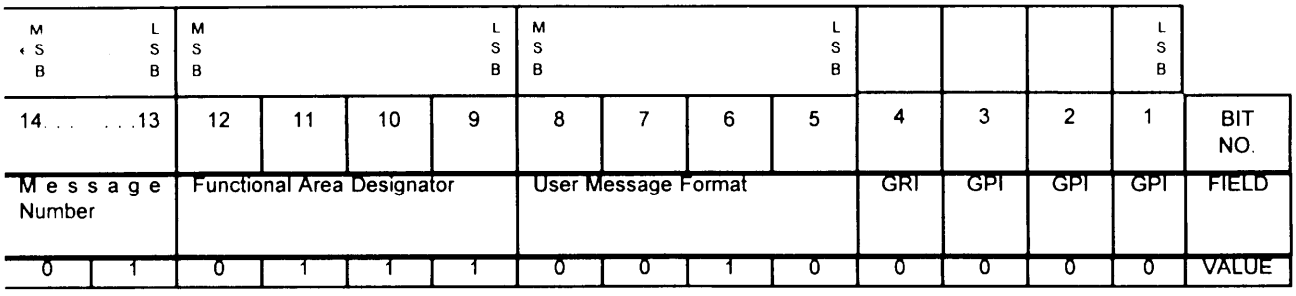


Figure 3. Application protocol data field bit order (example)

5.5.6.1 ASCII data element. In a data element composed of a string of 7-bit ASCII characters, the left most character shall be stored in memory first.

5.5.6.2 Binary data element. In a data element composed of a binary code, it shall be stored as a single data field.

5.5.6.3 Header format notations. The header format is depicted in Table I; the notations used to describe the header format are as follows:

a. Category will display an "M" for those fields that are mandatory. All other fields are conditional.

b. Group Code The group codes in Table I represent a logical grouping of information that is implemented as a "G" group. "G" groups within a header will be notated as GN were N indicates that numbered grouping (i.e., G1 indicates the first "G" group within the header: etc.). Nested groups are indicated by "GN/GN" notation where the left-most group is the highest level of the nesting and the right-most group is the current, lowest level.

c. Repeat codes. The repeat codes in Table I denote group appearance, nesting of groups, and maximum repetitions. The following notations are used:

(1)	R	=	Indicates this field is repeatable.
(2)	RN	=	Indicates this field is part of a group that can be repeated, with N specifying the group number (that is, R1 indicates the first repeatable group in the message).
(3)	C	=	Consecutive repeatability indicator specifies that the individual repeatable field or repeatable group of fields may be repeated only in a consecutive manner (for example, fields 3 and 4, followed immediately by fields 3 and 4). Consecutive repeating of a group of fields can occur only when the group is a single group (no nested groups) or the group is the innermost of several nested groups.
(4)	I	=	Iterative repeatability indicator specifies that the repeatable group may be repeated only after consideration has been given to all embedded iterative and consecutive repeatables.
(5)	(N)	=	Appears with the first field of a repeatable group, that is, R3C(4), and indicates the maximum number of appearances of the group in the message. The example, R3C(4), indicates the third repeatable group of the message that will be repeated consecutively and can appear a maximum of four times.
(6)	RN/RN	=	Nested repeating groups. Example R4/R5 R5 is nested in R4.

TABLE I. Application Header

Field Name	CAT	Group Code	Repeat Code	Description/ Resolution	Maximum Field Size (bits)
GPI	M	G1		ORIGINATOR ADDRESS GROUP	1
FPI		G1			1
UNIT REFERENCE NUMBER		G1			24
FPI		G1			1
UNIT NAME		G1			448
GPI	M			RECIPIENT ADDRESS GROUP	1
GRI			R1C(N) 0<N<=16		1
FPI			R1		1
UNIT REFERENCE NUMBER			R1		24
FPI			R1		1
UNIT NAME			R1		448
GPI	M			INFORMATION ADDRESS GROUP	1
GRI			R2C(16 - N)		1
FPI			R2		1
UNIT REFERENCE NUMBER			R2		24
FPI			R2		1
UNIT NAME			R2		448
GRI	M		R3C(N=16)	MESSAGE HANDLING GROUP	1
USER MESSAGE FORMAT	M		R3		4
FUNCTIONAL AREA DESIGNATOR	M		R3		4
MESSAGE NUMBER	M		R3		7
FPI	M		R3		1
MESSAGE SUBTYPE NUMBER			R3		7
FPI	M		R3		1
MESSAGE SIZE			R3		20
OPERATION INDICATOR	M		R3		2
RETRANSMIT INDICATOR	M		R3		1
MESSAGE PRECEDENCE CODE	M		R3		3
SECURITY CLASSIFICATION	M		R3		2
FPI	M		R3		1
CONTROL/RELEASE MARKING			R3		14

TABLE I. Application Header

Field Name	CAT	Group Code	Repeat Code	Description/ Resolution	Maximum Field Size (bits)
GPI	M		R3	ORIGINATOR DTG	1
YEAR		G3	R3		7
MONTH		G3	R3		4
DAY		G3	R3		5
HOUR		G3	R3		5
MINUTE		G3	R3		6
SECOND		G3	R3		6
GPI	M		R3	PERISHABILITY DTG	1
YEAR		G4	R3		7
MONTH		G4	R3		4
DAY		G4	R3		5
HOUR		G4	R3		5
MINUTE		G4	R3		6
SECOND		G4	R3		6
GPI	M		R3	ACKNOWLEDGMENT REQUEST GROUP	1
MACHINE ACKNOWLEDGE REQUEST INDICATOR		G5	R3		1
OPERATOR ACKNOWLEDGE REQUEST INDICATOR		G5	R3		1
OPERATOR REPLY REQUEST INDICATOR		G5	R3		1
GPI	M		R3	RESPONSE DATA GROUP	1
YEAR		G6	R3	DTG OF MESSAGE BEING ACKNOWLEDGED	7
MONTH		G6	R3		4
DAY		G6	R3		5
HOUR		G6	R3		5
MINUTE		G6	R3		6
SECOND		G6	R3		6
RECEIPT/COMPLIANCE		G6	R3	RESPONSE TO ACKNOWLEDGE REQUEST	3
FPI		G6	R3		1
CANTCO REASON CODE		G6	R3		3
FPI		G6	R3		1

TABLE I. Application Header

Field Name	CAT	Group Code	Repeat Code	Description/ Resolution	Maximum Field Size (bits)
CANTPRO REASON CODE		G6	R3		6
FPI		G6	R3		1
REPLY AMPLIFICATION		G6	R3		350

5.6 Application header fields

5.6.1 Originator, Recipient, and Information Addressee fields. These fields shall contain addresses that represent the names of the originating and receiving person(s) or process(es). The receiving application layer shall use the recipient and information fields to determine how the message shall be handled or delivered after the decoding process. The value in these fields depends on the person or process receiving the message. If a person is to be designated, the fields shall uniquely identify the individual so that the message may be routed to a specific mailbox or terminal. If a process is to be designated, these fields shall uniquely identify the process. The process must be associated with an end system to define the address uniquely. The following requirements apply to recipient and information addressee fields:

a. The recipient and information addressee fields shall be extendable to a combined total of 16 addressees.

b. When the recipient address is not present (GPI = 0), the message shall be broadcast in accordance with lower layer broadcast rules.

5.6.1.1 Unit Reference Number field. This field shall be a 24-bit binary code used by units on an interface to uniquely identify friendly military units. The applicable codes for this field depend on the setting of the User Message Format field (paragraph 5.6.2) and are specified in appendices to this document as referenced in Table II. The Unit Reference Number field and the Unit Name field are mutually exclusive fields (one or the other, not both).

5.6.1.2 Unit Name field. This field shall be a variable size field up to a maximum of 448 bits. It shall be in a character-coded format and used to uniquely identify a friendly military unit. This field is divided into 64 groups of 7 bits each representing an ANSI ASCII character. Special characters are legal. ASCII Delete (1111111) may be used as an end of text marker.

5.6.2 User Message Format field. This field shall be a 4-bit binary codeword representing the message formats shown in Table II. This field indicates the format of the message that is contained in the user data field and has association with the other message format-dependent fields, including, Functional Area Designator (paragraph 5.6.3), Message Number (paragraph 5.6.4), Message Subtype Number (paragraph 5.6.5), CANTCO Reason, (paragraph 5.6.19), and CANTPRO Reason (paragraph 5.6.20). The applicable codes for these fields are associated with the corresponding User Message Format in appendices to this document as shown in Table II.

TABLE II. User Message Format codes

Type of Message Format	Code	Message Format-Dependent Field/Code Reference
TADIL J (J-series message)	0000	TBD
UNDEFINED	0001	TBD
Variable Message Format (VMF) (K-series message)	0010	Appendix A
National Imagery Transmission Format System (NITFS)	0011	TBD
UNDEFINED	0100 - 1111	TBD

MSB LSB

5.6.3 Functional Area Designator field. This field shall contain a 4-bit binary codeword that identifies the functional area of a specific message using codewords. The Functional Area Designator combined with the Message Number field may be used to point to the applicable message of application processing. The applicable codes for this field depend on the setting of the User Message Format field (paragraph 5.6.2) and are specified in appendices to this document as referenced in Table II.

5.6.4 Message Number field. This field shall contain a 7-bit binary codeword that represents the number that identifies a specific message within a functional area (see paragraph 5.6.3). The Message Number value shall range from 1 to 127. The message set depends on the setting of the User Message Format field (paragraph 5.6.2) and is specified in appendices to this document as referenced in Table II.

5.6.5 Message Subtype Number field. This field shall contain a 7-bit binary codeword that represents the number that identifies a specific report within a message. The report set depends on the setting of the User Message Format field (paragraph 5.6.2) and is specified in appendices to this document as referenced in Table II.

5.6.6 Message Size field. This field shall contain a 20-bit binary number indicating the size, in bytes, of the associated message. Within the user data, messages which do not end on a byte boundary shall be zero-filled in the last byte starting with the first bit after the end of the message through the MSB of that byte. This field is required when there is more than one occurrence of the Message Handling Group (R3 in Table I).

5.6.7 Operation Indicator field. This field shall be a 2- bit binary codeword indicating the operational function of the message as shown in the Table III. used in support of either an operation, exercise, or simulation.

TABLE III. Operation indicator codewords

Operation indicator	Code
Operation	00
Exercise	01
Simulation	10
Undefined	11

MSB LSB

5.6.8 Retransmit Indicator field. This field shall be a 1-bit indicating whether a message is a retransmission. This field set to 1 shall affirm that the message is a retransmission. This field set to 0 shall indicate the negative.

5.6.9 Message Precedence field. This field shall be a 3-bit binary codeword indicating the relative precedence of a message as shown in Table IV.

TABLE IV. Message Precedence codewords

Precedence	Code	Explanation
Undefined	000-001	
Emergency command	010	Emergency Command
Undefined	011	
Flash	100	Reserved for initial entry contact messages of extreme urgency
Immediate	101	Reserved for messages relating to situations which gravely affect the security of forces
Priority	110	Reserved for messages requiring expeditious action by the addressees and/or furnishing essential information which routine will not handle
Routine	111	Used for all types of messages which justify transmission by rapid means unless of sufficient urgency to require a higher precedence

MSB LSB

5.6.10 Security Classification field. This field shall be a 2-bit codeword indicating the security classification of the message as shown in Table V.

TABLE V. Security Classification codewords

Classification	Code
Unclassified	00
Confidential	01
Secret	10
Top secret	11

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5.6.11 Control and Release Marking field. This field shall be a 14-bit binary field. It shall be encoded as two 7- bit ANSI ASCII characters indicating the restrictions or requirements for special handling, access control, and releasability of the message. Note: The complete list of data items for control and release marking is in DIA Manual 65-19, Standard Security Markings, as amended. The data items are classified confidential.

5.6.12 Originator Date and Time Group (DTG). These fields shall contain date and time information indicating the time that the message was prepared. This group combination shall be 34 bits long and shall contain data fields representing the year, month, day, hour, minute, and seconds of the message. Coding for each data field shall be as shown in Table VI.

TABLE VI. Date and Time Group codewords

ELEMENT	CODE	BITS
Year	0000000 - 1100011 (0) - (99)	7
Month	0001-1100 (1) - (12)	4
Day	00001 - 11111 (1) - (31)	5
Hour (24 hour clock)	00000 - 10111 (0000) - (2300)	5
Minute	000000 - 111011 (0) - (59)	6
Second	000000 - 111011 (0) - (59)	6

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5.6.13 Time Perishability DTG. The fields in this group provide the latest time the message is still of value. These fields shall be encoded as specified in paragraph 5.6.12.

5.6.14 Machine Acknowledge Request Indicator field. This field shall be a 1-bit binary codeword indicating whether the originator of a message requires a machine acknowledge for the message. This field set to 1 shall affirm that an acknowledgment is required. This field set to 0 shall indicate that a machine acknowledgment is not required.

5.6.15 Operator Acknowledge Request Indicator field. This field shall be a 1-bit binary codeword indicating whether the originator of a message requires an operator acknowledgment for the

message from the recipient. This field set to 1 shall affirm that an operator acknowledgment from the recipient is required. This field set to 0 shall indicate the negative.

5.6.16 Operator Reply Request Indicator field. This field shall be a 1-bit binary codeword indicating whether the originator of a message requires an operator reply to the message. This field set to 1 shall affirm that an operator reply to the message is required. This field set to 0 shall indicate the negative.

5.6.17 Message Acknowledgment DTG. The fields in this group provide the date and time of the original message that is being acknowledged. These fields shall be encoded as specified in 5.6.12.

5.6.18 Receipt/Compliance field. This field shall be a 3-bit binary codeword representing the receipt/compliance (R/C) codes shown in Table VII.

TABLE VII. Receipt/Compliance(R/C) codes

Type of Receipt/compliance	Code	Used by	Explanation
Undefined	000		
Machine Receipt [MR]	001	Recipient	Generated in response to a machine acknowledge request from the originator to indicate that the original message can be processed at the ultimate destination
Cannot Process [CANTPRO]	010	Recipient	Generated to indicate that an original message cannot be successfully processed at the ultimate destination
Operator Acknowledge [OPRACK]	011	Recipient	A positive operator-generated acknowledgment to indicate receipt of a message at the ultimate destination
Will Comply [WILCO]	100	Recipient	Generated to indicate that received message is understood and that the ultimate destination will comply
Have Complied [HAVCO]	101	Recipient	Generated to indicate that received message is understood and that the ultimate destination has complied
Cannot Comply [CANTCO]	110	Recipient	An operator generated response to indicate that a received message cannot or will not be carried out
Undefined	111		

MSB LSB

5.6.19 Cannot Comply (CANTCO) Reason field. This user-defined field shall be a 3-bit binary codeword indicating the reason that a recipient cannot comply with a particular message. The applicable codes for this field depend on the setting of the User Message Format field (paragraph 5.6.2) and are specified in appendices to this document as referenced in Table II.

5.6.20 Cannot Process (CANTPRO) Reason field. This user-defined field shall be a 6-bit binary codeword indicating the reason that a particular message cannot be processed by a recipient or information addressee. It shall be used only in R/C messages. The applicable codes for this field depend on the setting of the User Message Format field (paragraph 5.6.2) and are specified in appendices to this document as referenced in Table II.

5.6.21 Reply Amplification field. This field shall be a variable size up to a maximum of 350 bits. It shall be a character-coded field to provide textual data for an amplification of the recipient's reply to a message, if necessary. This field is divided into 50 groups of 7 bits each representing an ANSI ASCII character. Special characters are legal. ASCII Delete (1111111) may be used as an end of text marker.

5.7 Application header formatting rules and construction procedures. The case and condition syntax and procedures tabulated below shall be applied in the formatting and construction of the application header.

5.7.1 Case and conditionality statement syntax. The purpose of the case and conditionality statements is to rigorously and unambiguously define the construction rules for the application header so that it will be possible to achieve consistent construct implementations across multiple systems. They include cases for each use of the application header and the inter-element conditionalities within the application header for basic processing, defaults, legal entries, and special considerations.

5.7.1.1 Logical operators. Natural language does not lend itself to rigorous and unambiguous expression, therefore it is necessary to use well established logical operators to establish precise, mathematical meaning for logical relationships. The logical operators that will be used in this document are:

- AND - separates two discrete values and evaluates to true if both of the discrete values are true.
- OR - inclusive OR separates two discrete values and evaluates to true if at least one of the discrete conditions is true.
- XOR - exclusive OR separates two discrete values and evaluates to true if and only if one, not both, of the discrete conditions is true.
- NOT - a simple negation of the condition so that if A is true the NOT A would yield false.

The following truth table illustrates the meaning of the logical operator definitions given above. The table shows, for example, that given both "A" and "B" as true, then "NOT A" will yield false. "A AND B" will yield true, "A OR B" will yield true, and "A XOR B" will yield false. "A AND B" in this example represent names or action designators.

TABLE VIII. Logical Operator Definitions

A	B	NOT A	A AND B	A OR B	A XOR B
TRUE	TRUE	FALSE	TRUE	TRUE	FALSE
TRUE	FALSE	FALSE	FALSE	TRUE	TRUE
FALSE	TRUE	TRUE	FALSE	TRUE	TRUE
FALSE	FALSE	TRUE	FALSE	FALSE	FALSE

5.7.1.2 Application. Case and conditionality statements are used only to restrict the structure of the application header to a well-defined subset of the possible legal configurations that are specified by the application rules of application header construction.

5.7.1.3 Reserved words. Case statements reserved words that will be used in this document are:

- CASE - Identifies the title (purpose) under which the statement is defined.
- END CASE - Ends the case statement.
- IF..THEN..ELSE - Describes conditions under which statements are valid. The statement always starts with "IF" and must end with "ENDIF." An "IF" statement selects for execution, one or none of the enclosed sequence of statements depending on the (truth) value of one or more corresponding conditions.
- ELSIF - This keyword is used to extend the flexibility of the "IF..THEN..ELSE" construct. It is used when multiple conditions need to be evaluated in order to determine a logic path. Multiple "ELSIF" conditions are permitted. The general form is:

  IF *condition* THEN
  sequence of statements
  ELSIF *condition* THEN
  sequence of statements
  ELSE
  sequence of statements
  ENDIF
- ENDIF - Ends condition statement.

5.7.1.4 Cases. Case statements are a form of expressing a condition. The construct in this document indicates there must be at least two alternatives, and only one alternative at a time shall be valid. Case statements are used when a condition statement becomes too complex. A case statement may include an "XOR" (Exclusive OR) operator when it is possible to accomplish the same purpose in one or more ways. A case statement may also include an "OR" operator when any, or all, of several data elements can be used.

5.7.1.5 Conditions. Condition statements define the conditions under which a data group, data element, or value in a data element may be used. The condition statement is very structured in its use. The following is an example of the format of a conditional statement:

```
IF (condition)
  THEN
        (Sequence of Statements)
  ELSIF (condition)
  THEN (Sequence of Statements)
  ELSE (Sequence of Statements)
ENDIF
```

For the execution of an "IF" statement, the condition specified after "IF", and any conditions specified after other keywords are evaluated in succession until one evaluates to "TRUE", or all conditions are evaluated and yield "FALSE." If one condition evaluates to "TRUE", then the corresponding sequence of statements are executed; otherwise, none of the sequence of statements are executed.

5.7.1.6 Defaults. Defaults will be defined only if the receiving system's default value is of concern to the interface.

5.7.1.7 Expected response. The expected response by the system receiving an application header will depend on the content of the header fields and shall be stated as it relates to the case and conditionality statements for the header.

5.7.1.8 Special considerations. Special considerations cover those exceptions that cannot be defined under the preceding paragraphs.

5.7.1.9 Application header receipt. Upon receipt of an application header, a system shall validate the presence of all mandatory groups and fields, determine that all occurrence category conditions are satisfied, and validate the legality of all field entries to determine the legality of the header. This receipt processing is required for each header. It is the responsibility of the receiving system to apply default values, if specified, in those fields unspecified by the transmitting system.

5.7.2 Cases and conditions for the application header.

5.7.2.1 Cases.

5.7.2.1.1 Case: Message is an original message.

THIS CASE REQUIRES
GPI for Group 6 [Response Data Group] shall be "0" (Not Present)
AND Message body shall be present
END CASE

5.7.2.1.2 Case: Message is an acknowledgment message.

THIS CASE REQUIRES
GPI for Group 6 [Response Data Group] shall be "1" (Present)
AND GPI for Group 4 [Perishability DTG Group] shall be "0" (Not Present)
AND GPI for Group 5 [Acknowledgment Request Group] shall be "0" (Not Present)
AND Message body shall not be present
END CASE

5.7.2.2 Conditions.

5.7.2.2.1 Condition: IF the Originator Address Group is not present, THEN an acknowledgment shall not be requested.

IF GPI for G1 [ORIGINATOR ADDRESS GROUP] is set to "0" (Not Present)
THEN GPI for G5 [ACKNOWLEDGMENT REQUEST GROUP] shall be set to "0" (Not Present)
ENDIF

5.7.2.2.2 Condition: IF the Bit-coded Unit Reference Number is present, THEN the Character-coded Unit Name shall not be present in the same address group.

IF FPI for Unit Reference Number is set to "1" (Present)
THEN FPI for Unit Name shall be set to "0" (Not Present)
ENDIF

5.7.2.2.3 Condition: IF the Bit-coded Unit Reference Number is not present, THEN the Character-coded Unit Name shall be present in the same address group.

IF FPI for Unit Reference Number is set to "0" (Not Present)
THEN FPI for Unit Name shall be set to "1" (Present)
ENDIF

5.7.2.2.4 Condition: IF the Character-coded Unit Name is present, THEN the Bit-coded Unit Reference Number shall not be present in the same address group.

IF FPI for Unit Name is set to "1" (Present)
THEN FPI for Unit Reference Number shall be set to "0" (Not Present)
ENDIF

5.7.2.2.5 Condition: IF the Character-coded Unit Name is not present, THEN the Bit-coded Unit Reference Number shall be present in the same address group.

IF FPI for Unit Name is set to "0" (Not Present)
THEN FPI for Unit Reference Number shall be set to "1" (Present)
ENDIF

5.7.2.2.6 Condition: IF Recipient addressing is not present, THEN the message shall be a broadcast transmission.

IF GPI for R1 [Recipient Address Group] is set to "0" (Not Present)
THEN message shall be broadcast in accordance with lower layer broadcast protocols
ENDIF

5.7.2.2.7 Condition: IF Message Handling Group (R3) repeats, THEN Message Size shall be present.

IF there is more one occurrence of Message Handling Group (R3)
THEN FPI for Message Size shall be set to "1"
ENDIF

5.7.2.2.8 Condition: IF the message is not a CANTCO, THEN CANTCO Reason Code cannot be present.

IF Receipt/Compliance is NOT set to "6" (CANTCO)
THEN FPI for CANTCO Reason Code shall be set to "0" (Not Present)
ENDIF

5.7.2.2.9 Condition: IF the message is not a CANTPRO, THEN CANTPRO Reason Code cannot be present.

IF Receipt/Compliance is NOT set to "2" (CANTPRO)
THEN FPI for CANTPRO Reason Code shall be set to "0" (Not Present)
ENDIF

5.7.2.2.10 Condition: IF the message is not an Operator Reply, THEN Reply Amplification cannot be present.

IF Receipt/Compliance is NOT set to "4" (WILCO)
AND Receipt/Compliance is NOT set to "5" (HAVCO)
AND Receipt/Compliance is NOT set to "6" (CANTCO)
THEN FPI for Reply Amplification shall be set to "0" (Not Present)
ENDIF

5.7.2.3 Defaults. Default values for Message Precedence, Acknowledgments, and Message Classification shall be user defined.

5.7.3 Expected Response.

5.7.3.1 Machine Acknowledge Requested:

IF Machine Acknowledge Requested Indicator is set to"1"
THEN Response shall have Receipt/Compliance set to "1" (Machine Acknowledge)
OR Response shall have Receipt/Compliance set to "2" (CANTPRO)
ENDIF

5.7.3.2 Operator Acknowledge Requested:

IF Operator Acknowledge Requested Indicator is set to "1"
THEN Response shall have Receipt/Compliance set to "3" (Operator Acknowledge)
OR Response shall have Receipt/Compliance set to "2" (CANTPRO)
ENDIF

5.7.3.3 Operator Reply Requested:

IF Operator Reply Requested Indicator is set to "1"
THEN Response shall have Receipt/Compliance set to "4" (WILCO)
OR Response shall have Receipt/Compliance set to "5" (HAVCO)
OR Response shall have Receipt/Compliance set to "6" (CANTCO)
OR Response shall have Receipt/Compliance set to "2" (CANTPRO)
ENDIF

5.7.4 Special Considerations.

5.7.4.1 Perishable Data Check:

IF GPI for Group 4 [Perishable Data DTG] is set to "1"
AND Group 4 [Perishable Data DTG] is earlier than current DTG
THEN Message data shall be ignored
ENDIF

5.7.5 User data. This portion of the application protocol data unit shall contain the application process messages or data.

5.7.5.1 Message acknowledgments. A message acknowledgment is a report back to the originator on a receiving station's receipt of and intentions with respect to a received message. Acknowledgment requests are directed to message recipients only; they do not apply to information addressees. Acknowledgments are implemented in the acknowledgment header format.

5.7.5.2 Acknowledgment header format. Machine and operator acknowledgment request indicators are used by the originator to request a specific response from the receiving station, or appropriate operator, for selective acknowledgment of message receipt and compliance with the message instructions. A receiving station responds to the originator by sending an acknowledgment header. Depending on the type of acknowledgment request from the originator or the type of system involved, the response may be machine-generated (automatic) or operator-generated (manual) or a combination of both. The acknowledgment header consists of the following groups and fields (see also paragraph 5.7.2.1.2):

a. Acknowledgment originator address group (G1)

b. Acknowledgment recipient address group (R1)

c. Message handling group (R3). Within message handling group, the response data group (group 6), shall include the DTG of message being acknowledged and the R/C field.

5.7.5.3 Message accountability. The application header shall be used for the detection of duplicate messages and to associate an acknowledgment header with the original requesting message. The received fields of originator address group, originator date and time (DTG), functional area designator (FAD), and message number are used to uniquely identify a message. The originator shall guarantee the uniqueness of this combination of fields by ensuring that no original message is transmitted having the same DTG and FAD/message number.

a. Duplicate message check. The originator address group, originator DTG, FAD, and message number fields of each received message are compared with the corresponding fields of previous messages. Any duplicate messages (including retransmitted messages) shall be acknowledged if required and shall otherwise be ignored.

b. Acknowledgment matching. The originator address group, DTG of message being acknowledged, FAD, and message number fields of each received acknowledgment header are compared with the recipient address group, originator DTG, FAD, and message number fields of previously originated messages that require acknowledgment. The message handling application will maintain DTG, Originator Address, FAD, and Message Number information about previously received messages for a period of time long enough to exhaust the message originator's retransmission timers. Acknowledgment headers that match original messages shall be processed; unmatched Acknowledgment headers shall be ignored.

5.7.5.4 Message retransmission. A retransmission capability shall be provided to enable the automatic retransmission of a message that has not received an acknowledgment when one was requested.

a. The number of automatic retransmissions shall be selectable with a range of 0 to 3. The parameter governing the number of retransmissions shall be separately selectable for each FAD/Message Number combination.

b. A timer shall be provided to schedule the automatic retransmission. Expiration time shall be selectable with a range of 5 to 600 seconds. Upon expiration of the timer, provided an acknowledgment has not been received, the message shall be retransmitted by the originating system. If an acknowledgment is not received prior to expiration of the timer on the final retransmission, the operator shall be notified. Messages containing perishable data and requiring acknowledgment shall have the Perishability DTG set to a time later than the retransmit timeout.

5.8 Processing Factors.

5.8.1 Application process. The application process shall provide the application layer the bit-oriented or character-oriented messages that satisfy information exchange requirements.

5.8.2 Message formats. The message formats shall be user-defined. The user message indicator field in the application layer header provides a flag that specifies the message format that is being used in the application process.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Subject term (key word) listing. The following key words and phrases apply to this MIL-STD.

Application Header
Application Layer
MIL-STD-188-220
Intranet
Selective Directed Broadcast
IF
THEN
CASE
CANTCO
CANTPRO
Message
VMF TIDP

6.2 Issue of the DOD Index of Specifications and Standards When this MIL-STD is used in procurement, the applicable issue of the DODISS must be cited in the solicitation.

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APPENDIX A

APPLICATION HEADER FIELDS AND CODES FOR VARIABLE MESSAGE FORMATS

A.1 General

A.1.1 Scope. This appendix contains definition of the Variable Message Format (VMF) codes and values for application header fields that are dependent on the setting of the User Message Format field.

A.1.2 Application. This appendix is conditional based on the setting of the User Message Format field as indicated in paragraph 5.6.2 and Table II of this standard. If the User Message Format field is set to "2", this appendix is mandatory for application headers pertaining to VMF messages. For all other settings of User Message Format field, this appendix is optional.

A.2 Applicable Documents.

GOVERNMENT STANDARDS

None

OTHER GOVERNMENT DOCUMENTS

Joint Interoperability of Tactical Command and Control Systems, Variable Message Format Technical Interface Design Plan (Test Edition) (VMF TIDP TE)

VOL I *VMF Overview*

VOL II *Data Element Dictionary*

VOL III *K-Series Message Formats*

Variable Message Format Interface Operating Procedures (VMF IOP)

A.3 Codeword Tables.

A.3.1 Unit Reference Number codewords. The VMF codes for the Unit Reference Number field shall be assigned in accordance with the VMF IOP.

A.3.2 Functional Area Designator codewords. The VMF codes for the Functional Area Designator (FAD) field are defined in Table A-1. The FAD field is defined in paragraph 5.6.3 of this document.

TABLE A-1. Functional Area Designator codewords

Functional Area	Code
Network Control	0000
System Information Exchange	0001
Fire Support	0010
Air Operations	0011
Intelligence Operations	0100
Land Combat Operations	0101
Maritime Operations	0110
Combat Service Support	0111
Special Operations	1000
JTF Operations Control	1001
Air Defense/Air Space Control	1010
Undefined	1011-1111

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A.3.3 Message Number codewords. The VMF codes for the Message Number field are listed in Volume III of the Joint VMF TIDP. The Message Number field is defined in paragraph 5.6.4 of this document.

A.3.4 Message Subtype Number codewords. The VMF codes for the Message Subtype Number field are listed in Volume II of the Joint VMF TIDP, Data Unit Identifier (DUI) REPORT/MESSAGE/OVERLAY TYPE. The Message Subtype Number field is defined in paragraph 5.6.5 of this document.

A.3.5 CANTCO Reason codewords. The VMF codes for the CANTCO Reason field are defined in Table A-2. The CANTCO Reason field is defined in paragraph 5.6.19 of this document.

TABLE A-2. CANTCO Reason codewords

CANTCO reason	Code
Communications problem	000
Ammunition problem	001
Personnel problem	010
Fuel problem	011
Terrain/Environment problem	100
Equipment problem	101
Tactical Situation problem	110
Other	111

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A.3.6 CANTPRO Reason codewords. The VMF codes for the CANTPRO Reason field are defined in Table A-3. The CANTPRO Reason field is defined in paragraph 5.6.20 of this document.

TABLE A-3. CANTPRO Reason codewords

CANTPRO Reason	Code
Undefined	000000
CANTPRO CODE 1 - Field content invalid	000001
CANTPRO CODE 2 - Message incorrectly routed	000010
CANTPRO CODE 3 - Address inactive	000011
CANTPRO CODE 4 - Reference point unknown to receiving agency	000100
CANTPRO CODE 5 - Fire units must be controlled by receiving agency	000101
CANTPRO CODE 6 - Mission must be controlled by receiving agency	000110
CANTPRO CODE 7 - Mission number unknown by receiving agency	000111
CANTPRO CODE 8 - Target number unknown by receiving agency	001000
CANTPRO CODE 9 - Schedule number unknown by receiving agency	001001
CANTPRO CODE 10 - Incorrect controlling address for a given track number	001010
CANTPRO CODE 11 - Track number not in own track file	001011
CANTPRO CODE 12 - Invalid according to given field	001100
CANTPRO CODE 13 - Message cannot be converted	001101
CANTPRO CODE 14 - Agency file full	001110
CANTPRO CODE 15 - Agency does not recognize the message number	001111
CANTPRO CODE 16 - Agency cannot correlate message to current file content	010000
CANTPRO CODE 17 - Agency limit exceeded on repeated fields or groups	010001
CANTPRO CODE 18 - Agency computer system inactive	010010
CANTPRO CODE 19 - Addressee unknown	010011
CANTPRO CODE 20 - Can't forward (agency failure)	010100
CANTPRO CODE 21 - Can't forward (link failure)	010101
CANTPRO CODE 22 - Illogical juxtaposition of header fields	010110
Undefined	010111-111111

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Concluding Material

a. Preparing Activity:

Defense Information Systems Agency (DISA) - DC

b. Custodians:

DISA:	DC
Army:	SC
Air Force:	90
Navy:	OM
DIA:	DI
NSA:	NS
USMC:	MC
DLA:	DH
Other:	Joint Staff/Architecture & Integration
	USSPACECOM

c. Review Activities:

Army	SC, PT
Air Force	3, 17, 29, 33, 90
DLA	DH
DMA	MP
DIA	DI
DOT	OST
NSA	NS
OASD	IQ, DO, MA, IR
ODISC4	AC
NAVY	EC, CH, ND, TD, OM
USMC	MC, CG

d. Project number:

DCPS-0020, SP 01

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1,2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1 DOCUMENT NUMBER MIL-STD 2045-47001	2 DOCUMENT DATE (YYMMDD) 950727

3 DOCUMENT TITLE: Interoperability Standard for Connectionless Data Transfer Application Layer Standard

4 NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b ORGANIZATION	
c ADDRESS (Include Zip Code)	d TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8 PREPARING ACTIVITY

a NAME Frank Curcio	b TELEPHONE (Include Area Code) (1) Commercial (908)532-7732 (2) AUTOVON 992-7732
c ADDRESS (Include Zip Code) ATTN: JEBBF Building 283 JIEO Ft. Monmouth, NJ 07703-5613	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340